Options III: Trading Strategies

BUSS386. Futures and Options

Prof. Ji-Woong Chung

Lecture Outline

- Three option products in the headlines — and why each is a combination of Lecture 6's four payoff diagrams
- Bond + option: the principal-protected note, and its mirror image, the ELS
- Stock + option: covered call, protective put, collar — insurance versus hedging
- Spreads: bull, bear, butterfly, box, calendar — betting on direction, or on *calm*
- Combinations: straddle and strangle — betting on a *move* without picking a direction
- Matching the strategy to the view: direction $\times$ volatility

Three option products in the headlines

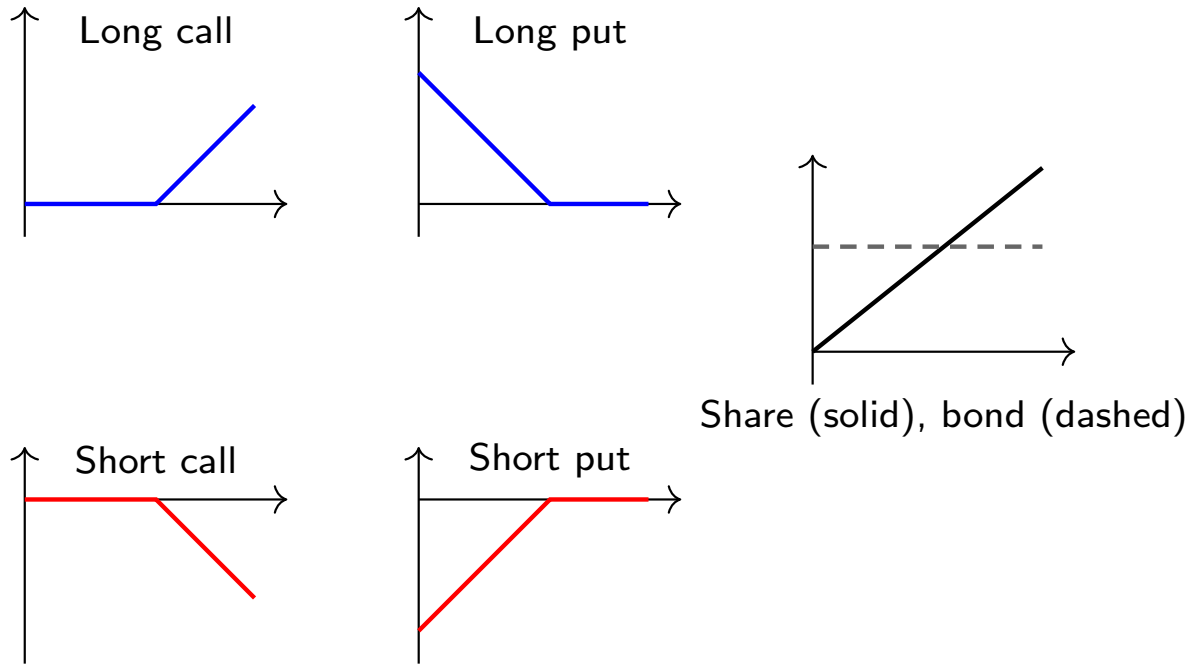
- **ELS** (equity-linked securities), sold at Korean bank branches as a "deposit with a bonus." Of the ₩6.66tn of Hang Seng China Enterprises Index (HSCEI)-linked ELS that had matured by mid-May 2024, ₩3.16tn was lost — an average loss of 47.4%. [Korea Herald, 21 May 2024]

- **Covered-call ETFs**, sold as "monthly dividend" funds. KRX-listed covered-call ETFs held ₩28.9tn on 28 Sep 2026, more than four times the level two years earlier. [The Standard, 21 Oct 2026]

- **0DTE options** in the US, expiring the day they are bought. In 2025, 59% of all S&P 500 index option volume — 2.3 million contracts a day — was 0DTE. Not available on KRX, where the shortest KOSPI 200 option is a weekly. [Cboe, 6 Jan 2026]

Every one of these is a **bond or a share plus one or two options**. Today's job: learn to read the combination, and to know which view it expresses.

Building blocks — the four positions, plus a share and a bond



- Every strategy today is a **sum** of these six pictures.
- Put-call parity (Lecture 7) is the glue.

Synthetic positions — one identity, many uses

From put-call parity for European options on a non-dividend stock,

$$c_0 - p_0 \;=\; S_0 - Ke^{-rT}.$$

Rearrange to manufacture whichever leg you cannot trade directly:

Want to be	Trade instead
Long stock	long call + short put + lend Ke^{-rT}
Short stock	short call + long put + borrow Ke^{-rT}
Long call	long stock + long put + borrow Ke^{-rT}
Long put	long call + short stock + lend Ke^{-rT}

- When short-selling is restricted — as on KRX for most of 2023–2025 — a synthetic short via options is the workaround.

Bond + Option

Principal-protected note — the idea

- A note that returns the **full principal** at maturity, plus a bonus if the market rises. Investors who "cannot afford to lose" love it.
- Recipe: a **zero-coupon bond** that grows to the principal, plus a **call option** bought with whatever money is left over.
- In Korea the same structure is sold as an **ELB** (equity-linked bond), usually on the KOSPI 200.

e.g. A Korean securities firm offers a 1-year ELB: *"If the KOSPI 200 at maturity is not below its level at inception, the note pays 3.0% per annum at minimum. Principal is guaranteed either way."*

Principal-protected note — the construction

You invest ₩10,000,000 for one year; $r = 3\%$ (c.c.). The firm:

1. Buys a 1-year zero-coupon bond with face value ₩10,000,000. It costs

$$10{,}000{,}000\, e^{-0.03} = 9{,}704{,}455, \qquad \text{leaving } 295{,}545.$$

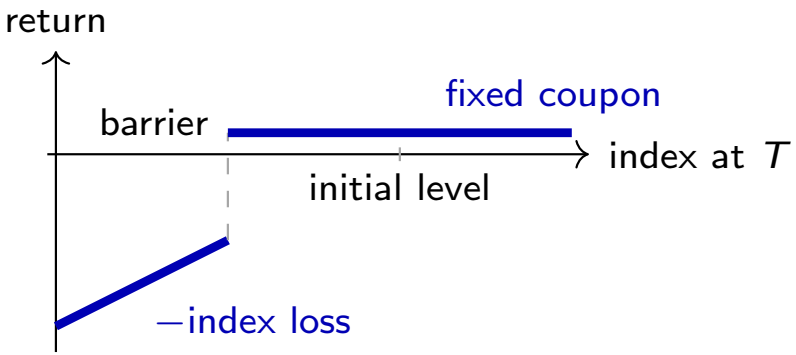
2. Spends the leftover on a 1-year **European call** on the KOSPI 200 struck at today's level S_0. A call on ₩10,000,000 of index costs, say, ₩850,000, so the firm can afford $280{,}500/850{,}000 = 0.33$ of one and keeps ₩15,045 as its fee.

Value at maturity:

$$\underbrace{10{,}000{,}000}_{\text{bond}} + \underbrace{0.33 \times 10{,}000{,}000 \times \max\left(\frac{S_T}{S_0} - 1,\, 0\right)}_{0.33 \text{ of a call}}.$$

- Principal back for sure, plus a **participation rate** of 33% in any index gain.

The ELS — same ingredients, opposite side of the option



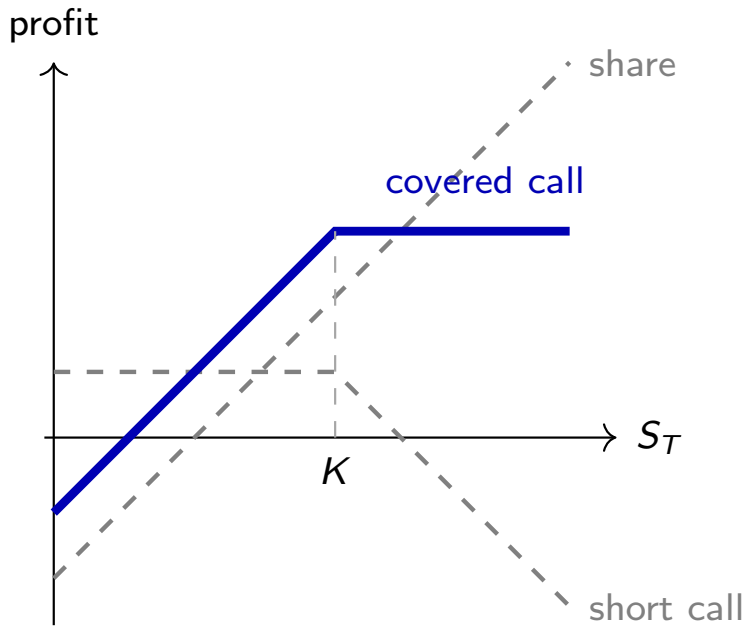
- Bond + **short** put (with a *barrier*, typically 50–65% of the starting level). The "coupon" is the put premium the saver is paid.
- While the index stays above the barrier, the put expires worthless and the saver keeps the coupon. Below it, the saver eats the index loss one for one.

- HSCEI-linked ELS sold in 2021 near 12,000; the index halved by late 2023, breaching the barriers.
- Lesson: **premium received is never free**. It is the price of a tail you have agreed to carry.

Stock + Option

Writing a covered call

Position. Long 1 share + short 1 call (strike K, premium c).



- You collect c today and give up every dollar above K. Income on a share you already own and are willing to part with at K.
- The profit line has the shape of a **short put**. That is not a coincidence.

Covered call = short put, by parity

Rearrange put-call parity:

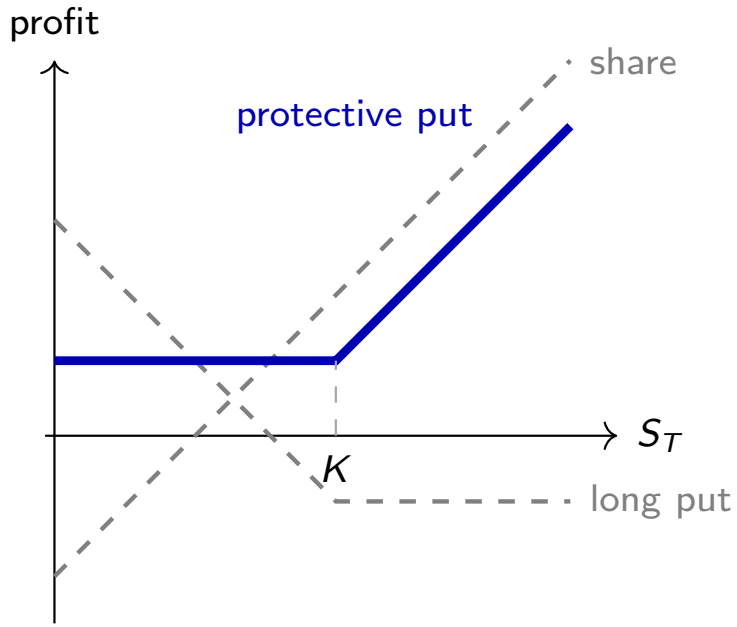
$$\underbrace{S_0 - c_0}_{\text{covered call}} = \underbrace{Ke^{-rT} - p_0}_{\text{lend } Ke^{-rT} \text{ and sell a put}} .$$

Same cost today, same payoff at T — so the same profit diagram.

- Writing covered calls is *selling volatility*: you earn the premium when the stock stays quiet and lose the upside when it does not.
- **The products.** "Monthly dividend" covered-call ETFs — JEPI and QYLD in the US, KODEX and TIGER covered-call ETFs on KRX (₩28.9tn by Sep 2026) — are exactly this.
- **The fine print.** The "dividend" is option premium. In a strong rally the fund lags the index by construction; in a crash it falls with it, less one month of premium.

Protective put

Position. Long 1 share + long 1 put (strike K, premium p).



- **Insurance.** The downside is floored at $K - p$; the upside is kept, less the premium.
- The put premium is the insurance cost, and $S_0 - K$ is the deductible you retain. A lower strike is cheaper insurance with a bigger deductible.
- By parity, share + put = call + bond: a protective put is a **long call**.

Insure with a put, or hedge with a forward?

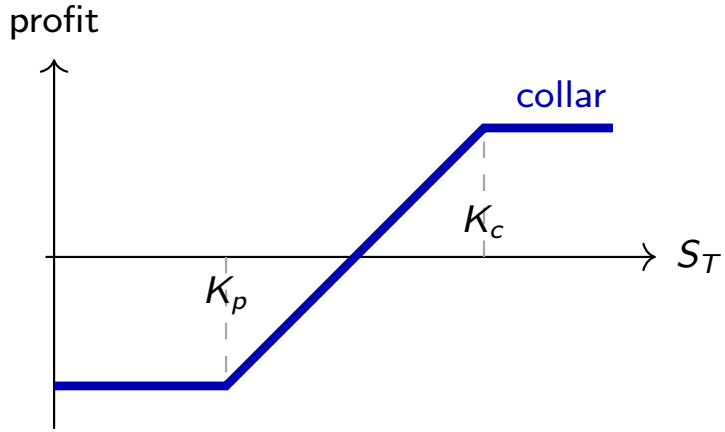
Setup. A fund manager is long a stock at S_0 and fears a fall. Two ways out:

	Short forward at K	Long put at K
Cost today	zero	premium p
If S_T falls	locked at K	floor at K
If S_T rises	*still* locked at K	keep the upside
Shape	flat line	kinked (one-sided)

- The forward is free but **symmetric**: it removes the good outcomes with the bad ones.
- The put costs money and is **asymmetric**: you pay to keep the right-hand side of the picture.

Collar — insurance that pays for itself

Position. Long 1 share + long put at K_p + short call at K_c, with $K_p < S_0 < K_c$.



- Sell the upside above K_c to pay for the floor at K_p. If the call premium equals the put premium, it is a **zero-cost collar**.
- Used by founders and executives hedging a concentrated stock position without selling it, and by "hedged equity" ETFs that roll a collar on the S&P 500 every quarter.
- Price of the insurance: both tails are gone. You hold a stock that can move only between K_p and K_c.

Spreads

Spreads — what they are and why traders like them

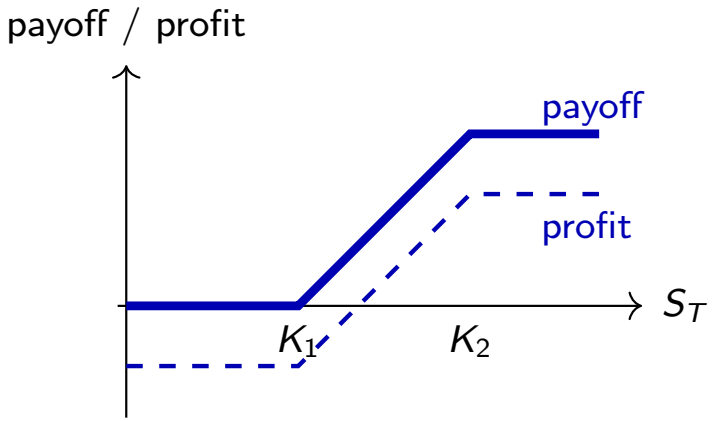
A **spread** combines two or more options of the *same type* (all calls, or all puts) on the same underlying.

- Different strikes, same expiry $\rightarrow$ **vertical** spread (bull, bear, butterfly).
- Same strike, different expiries $\rightarrow$ **calendar** (horizontal) spread.
- Both differ $\rightarrow$ **diagonal** spread.

A spread has a **known maximum gain and maximum loss**. That caps the margin a broker demands, which is why spreads — not naked options — dominate retail option trading.

Bull spread

Position. Long call at K_1 + short call at K_2, $K_1 < K_2$, same expiry. View: *moderately* bullish.

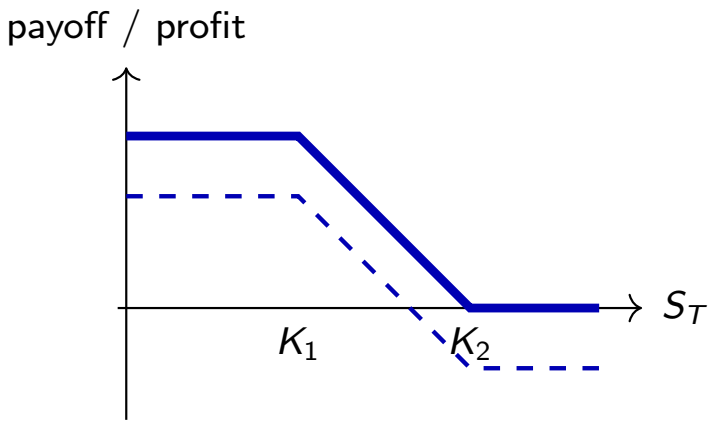


	$S_T < K_1$	$K_1 \le S_T < K_2$	$S_T \ge K_2$
Long K_1 call	0	$S_T - K_1$	$S_T - K_1$
Short K_2 call	0	0	$-(S_T - K_2)$
Payoff	0	$S_T - K_1$	$K_2 - K_1$

- The K_1 call costs more than the K_2 call, so you pay a net premium $\pi = c(K_1) - c(K_2) > 0$. Max loss π; max gain $K_2 - K_1 - \pi$.
- You sold the upside above K_2 to make the bet cheaper.
- Same shape with puts (long K_1 put, short K_2 put), but then you *receive* premium up front.

Bear spread

Position. Long put at K_2 + short put at K_1, $K_1 < K_2$, same expiry. View: *moderately* bearish.



	$S_T < K_1$	$K_1 \le S_T < K_2$	$S_T \ge K_2$
Long K_2 put	$K_2 - S_T$	$K_2 - S_T$	0
Short K_1 put	$-(K_1 - S_T)$	0	0
Payoff	$K_2 - K_1$	$K_2 - S_T$	0

- The K_2 put costs more, so again a net premium is paid. Max gain $K_2 - K_1 - \pi$; max loss π.
- **Mirror rule.** Bull = buy the low strike, sell the high strike. Bear = buy the high strike, sell the low strike.

Example — a bear spread built from calls

Q. An investor expects the S&P 500 to fall. Two 1-year European calls on the index are quoted: $K = 6{,}000$ at \$320 and $K = 6{,}200$ at \$200. Which does she buy, which does she sell, and what is her profit at expiry as a function of S_T? Where does she break even?

Example — a bear spread built from calls

Q. Calls: $K = 6{,}000$ at \$320, $K = 6{,}200$ at \$200. Bear spread?

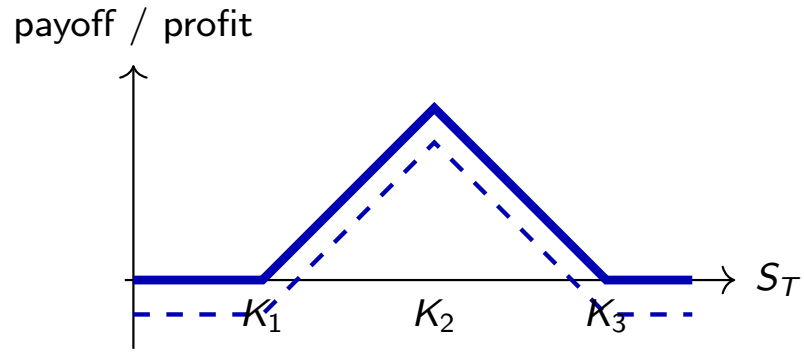
A. Bear = buy the *high* strike, sell the *low* strike: long the 6,200 call, short the 6,000 call. Cash today: $+320 - 200 = \boxed{+\$120}$ — a bear spread with calls is a **credit** trade.

	$S_T < 6{,}000$	$6{,}000 \le S_T < 6{,}200$	$S_T \ge 6{,}200$
Short 6,000 call	0	$-(S_T - 6{,}000)$	$-(S_T - 6{,}000)$
Long 6,200 call	0	0	$S_T - 6{,}200$
Payoff	0	$6{,}000 - S_T$	-200
Profit	$+120$	$6{,}120 - S_T$	$\boxed{-80}$

- Max profit \$120 (the credit, kept if $S_T < 6{,}000$). Max loss \$80 $= 200 - 120$. Breakeven $S_T = 6{,}120$.
- Same picture as the put version; only the timing of the cash differs. Parity, once more.

Butterfly spread — betting that nothing happens

Position. Long K_1 call + long K_3 call + short two K_2 calls, $K_2 = \frac{1}{2}(K_1 + K_3)$.



- Pays $K_2 - K_1$ if S_T lands exactly on K_2, zero at either wing. Max loss is the (small) premium paid.
- The view is not about direction. It is a bet on **low volatility**: "the stock will still be around K_2 at expiry."
- Because the payoff is never negative, the butterfly must cost something: $c(K_1) - 2c(K_2) + c(K_3) > 0$. Call prices are *convex* in the strike — a free-lunch check you can run on any option chain.

Butterfly spread — payoff table

- Payoff at expiry, with $K_2 = \frac{1}{2}(K_1 + K_3)$:

	$S_T < K_1$	$K_1 \le S_T < K_2$	$K_2 \le S_T < K_3$	$S_T \ge K_3$
Long K_1 call	0	$S_T - K_1$	$S_T - K_1$	$S_T - K_1$
Long K_3 call	0	0	0	$S_T - K_3$
Short two K_2 calls	0	0	$-2(S_T - K_2)$	$-2(S_T - K_2)$
Payoff	0	$S_T - K_1$	$\underbrace{2K_2 - K_1 - S_T}_{=K_3 - S_T}$	$\underbrace{2K_2 - K_1 - K_3}_{=0}$

- The same tent can be built from puts: long K_1 put, long K_3 put, short two K_2 puts.

- Reverse every sign and you have a **short butterfly**: a cheap bet that the stock will *leave* the range $[K_1, K_3]$.

Box spread — a loan made of options

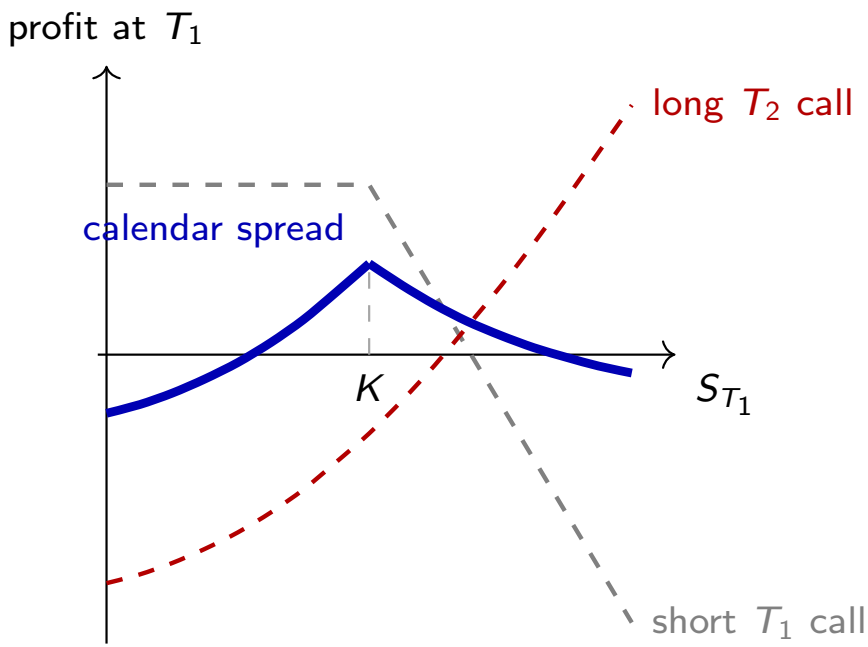
Position. Bull spread with calls + bear spread with puts, same $K_1 < K_2$ and expiry.

	$S_T < K_1$	$K_1 \le S_T < K_2$	$S_T \ge K_2$
Bull call spread	0	$S_T - K_1$	$K_2 - K_1$
Bear put spread	$K_2 - K_1$	$K_2 - S_T$	0
Payoff	$K_2 - K_1$	$K_2 - K_1$	$K_2 - K_1$

- The payoff is $K_2 - K_1$ **whatever happens**. Four options have manufactured a zero-coupon bond.
- No-arbitrage price today: $(K_2 - K_1)e^{-rT}$. If the four legs cost less, buy the box and earn more than r; if more, sell the box and borrow below r.
- Traders use S&P 500 index boxes as a financing market. The rate implied by the box is a clean test of whether four option quotes are mutually consistent.

Calendar spread — same strike, different expiries

Position. Short a call expiring at T_1 + long a call with the same K expiring at $T_2 > T_1$.



$K = 100$, 6-month vs. 1-year calls, $r = 3\%$. Net premium paid $11.35 - 7.76 = 3.59$.

- At T_1 the T_1 call is *settled* (kinked gray line). The T_2 call is still alive and must be *valued*, not paid off (smooth red curve). The sum has a corner at K and curves everywhere else.
- Profit peaks at $S_{T_1} = K$: the T_1 call dies worthless while the T_2 call still carries its maximum time value.
- A bet on a **quiet market**, like the butterfly.

Calendar spread — the cash flows

- At $t = 0$: receive $c_0(T_1)$, pay $c_0(T_2)$. Net $c_0(T_1) - c_0(T_2) < 0$.
- At T_1: the short call settles and the long call is sold:

$$-\max(S_{T_1} - K, 0) \; + \; c_{T_1}(T_2),$$

  where $c_{T_1}(T_2)$ is the market price at T_1 of a call with $T_2 - T_1$ left to run.

- We cannot price $c_{T_1}(T_2)$ yet — that needs next week's model — but Lecture 7 already **bounds** it:

$$\max\left(S_{T_1} - Ke^{-r(T_2 - T_1)}, 0\right) \; \leq \; c_{T_1}(T_2) \; \leq \; S_{T_1}.$$

Example — closing a calendar spread

Q. In November 2025 an investor sold a 1-year European call and bought a 2-year European call, both with $K = \$120$ on the same non-dividend stock. The 1-year call fetched \$5 and the 2-year call cost \$8. It is now November 2026, the stock trades at \$150, and $r = 3\%$ (c.c.). The investor closes everything today. What are the smallest and largest possible profits?

Example — closing a calendar spread

A. Cash flow today (Nov 2026):

$$-\max(150-120,\,0)+c_t(\text{Nov 2027}) \;=\; -30+c_t(\text{Nov 2027}).$$

- The remaining call has one year left, so its price must satisfy

$$\underbrace{\max(150-120e^{-0.03},\,0)}_{=33.547} \;\le\; c_t(\text{Nov 2027}) \;\le\; 150.$$

- Add the initial net premium ($+5-8=-3$, ignoring interest):

$$0.547 \;\le\; \underbrace{-30+c_t(\text{Nov 2027})-3}_{\text{profit}} \;\le\; 117.$$

- The lower bound is what matters in practice: with the stock deep in the money, the long call is nearly all intrinsic value and the spread earns only $K(1-e^{-r})-3\approx 0.55$. The "calm market" bet did not pay because the market was not calm.

Calendar or butterfly? Two ways to bet on a quiet market

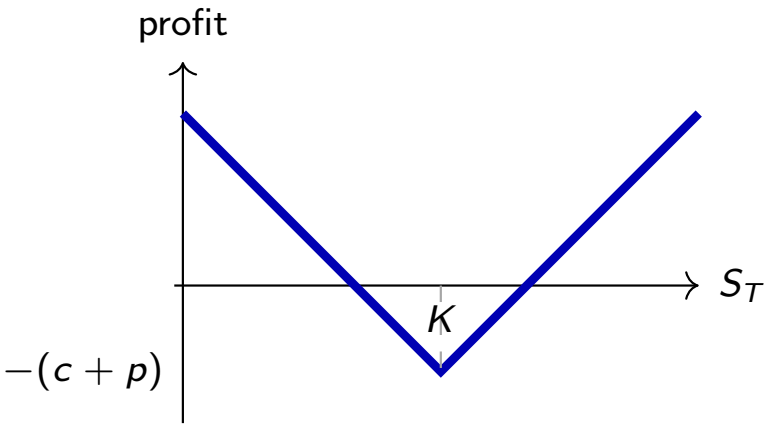
	Butterfly	**Calendar**
Legs	3 strikes, 1 expiry	1 strike, 2 expiries
Payoff when you exit	Known in advance: the tent in S_T. No model needed.	Depends on what the remaining call *sells for* at T_1.
What else moves it	Nothing at expiry: only S_T	The long leg gains if the market turns nervous, loses if it calms further
Worst case	Premium paid, at either wing	About the premium paid
Natural use	"Stock near K_2 on a date" with no volatility exposure	Near-dated options look expensive relative to later ones: earnings, a rate decision, an election

- Both profit if the stock sits still, but the butterfly bets on the **price** alone, while the calendar also bets that **short-dated volatility deflates faster than long-dated** once the event passes.
- Liquidity and trading costs: a calendar is two trades, a butterfly is four.

Combinations: Straddle and Strangle

Straddle — betting on a move, not a direction

Position. Long call + long put, same K, same T. A *combination* mixes calls and puts.

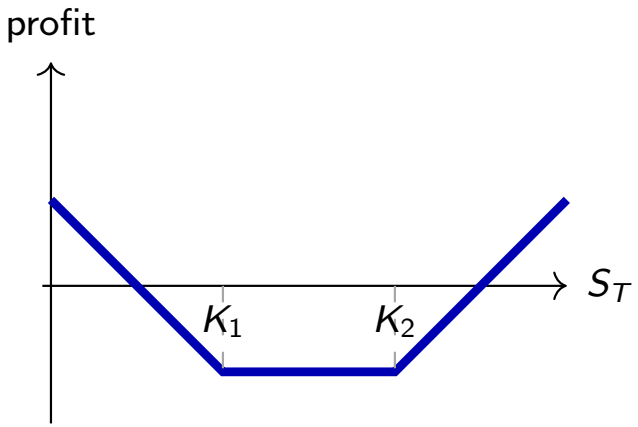


	$S_T < K$	$S_T \geq K$
Long call	0	$S_T - K$
Long put	$K - S_T$	0
Payoff	$K - S_T$	$S_T - K$

- Payoff $= |S_T - K|$. You profit if the stock moves **more than $c + p$ in either direction**.
- Natural around scheduled news: earnings, an election, a central-bank meeting. The straddle price *is* the market's forecast of the size of the move — if you expect more, buy it.

Strangle — the cheaper straddle

Position. Long put at K_1 + long call at K_2, $K_1 < K_2$, same T. Both legs out of the money.



	$S_T < K_1$	$K_1 \le S_T < K_2$	$S_T \ge K_2$
Long K_2 call	0	0	$S_T - K_2$
Long K_1 put	$K_1 - S_T$	0	0
Payoff	$K_1 - S_T$	0	$S_T - K_2$

- Cheaper than the straddle because both options are out of the money — but the stock must now travel *further* before you profit.
- The trade-off every option buyer faces: lower premium, lower probability of payout.

Example — where does a strangle make money?

Q. An investor buys a European call with $K = \$80$ for \$6 and a European put with $K = \$50$ for \$5, both expiring at T. For which values of S_T is the profit positive?

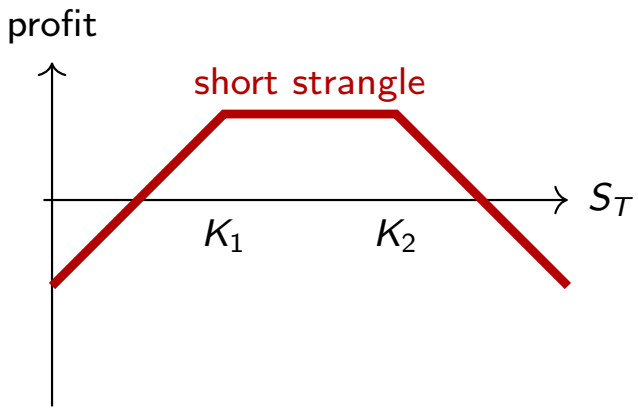
Example — where does a strangle make money?

A. Total premium paid: $6 + 5 = \$11$.

	$S_T < 50$	$50 \le S_T < 80$	$S_T \ge 80$
Long 80 call	0	0	$S_T - 80$
Long 50 put	$50 - S_T$	0	0
Payoff	$50 - S_T$	0	$S_T - 80$
Premium	-11	-11	-11
Profit	$39 - S_T$	-11	$S_T - 91$

- Profit > 0 when $S_T < 39$ or $S_T > 91$. **Breakevens: 39 and 91.**
- Between 50 and 80 the full \$11 is lost. The wide gap between the strikes is exactly what made the strangle cheap.

The other side — selling straddles and strangles



- Flip every sign: collect $c + p$ today, keep it if the market stays inside the strikes.
- Profit is **capped** at the premium; loss is **unlimited** on the upside and K_1 on the downside.

- This is the engine of "income" option strategies, including most systematic 0DTE programs: a positive return most days, and an occasional day that erases weeks of gains. "Picking up nickels in front of a steamroller."
- The ELS is a short put with a barrier. The covered call is a short call. The short strangle is both at once.

Which strategy for which view?

	Expect a big move	**Expect calm**
Bullish	long call	bull spread; covered call (if you own the stock)
No view on direction	straddle, strangle	butterfly, calendar, short straddle or strangle
Bearish	long put	bear spread
Own the stock, want a floor	protective put	collar
Want a bond		box spread

- Everything in the right-hand column *sells* volatility and earns premium; everything in the left-hand column *buys* it and pays premium. Next week we start to put a number on that premium.

Summary — what to take away

- Every strategy is a **sum of the six building blocks**: four option positions, a share, a bond. Draw the legs and add them vertically.
- **Parity** makes different-looking positions identical: covered call = short put, protective put = long call, bull spread with calls = bull spread with puts.
- Spreads trade **direction with a cap**; butterflies and calendars bet on **calm**; straddles and strangles bet on **movement**.
- **Premium received is a tail you now carry** — the covered-call ETF, the ELS, and the short strangle differ only in packaging.
- A strategy is chosen by two views: **which way** and **how much**. The second is the volatility view, and it is the heart of option pricing.

Next week: our first pricing model — the binomial tree. One step, then many; risk-neutral probabilities; and why the replicating portfolio, not the expected return, sets the price. Reading: Hull, Ch. 13.

Appendix: 0DTE Options

What is a 0DTE option?

- "Zero days to expiration": an option that expires on the day it is traded.
- Possible because Cboe now lists S&P 500 index (SPX) options expiring **every weekday** (Monday and Wednesday expiries were added in 2022, completing the set); SPY and QQQ followed.
- SPX options are European and cash-settled, so a 0DTE position is closed by 4 p.m. New York time with no overnight risk and no assignment.
- Scale: 0DTE was 59% of SPX volume in 2025, 2.3 million contracts a day, up from well under half before 2022. [Cboe, 6 Jan 2026]

Who trades 0DTE, and with which of today's strategies

- **Retail:** long calls or puts around scheduled events (Fed days, jobs reports). Cheap, lottery-like, usually worthless by the close.
- **Income programs:** short strangles and **iron condors** — a short strangle with far wings bought back for protection, i.e. a bear put spread plus a bull call spread sold together. Credit each day; tail days hurt.
- **Dealers and hedge funds:** hedging the other side, and trading intraday volatility.
- Evidence on the retail cohort: using Cboe trade data from 2021–2023, Beckmeyer, Branger and Gayda find retail 0DTE traders lost about $358,000 per day in aggregate. [SSRN 4404704, 2023]

Nothing in a 0DTE book is new: it is today's spreads and combinations with the clock set to one day. What changes is that **time value** vanishes within hours — the subject of Lecture 11.